BECS 22623
Implementation of Numerical Methods using Appropriate software

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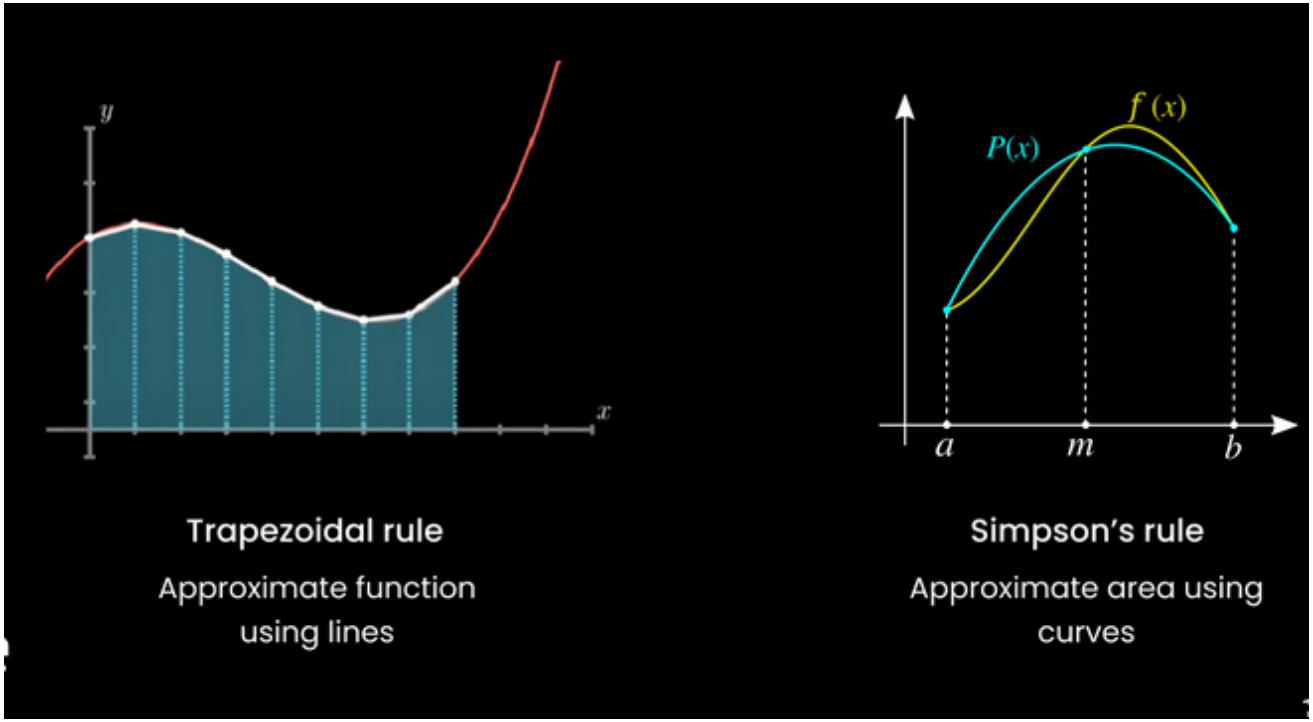
Outline

- Mid Point Rule
- Trapezoidal Rule
- Simpson's 1/3 Rule

Numerical Integration



Mid Point Rule

Let $f(x)$ be defined on a closed interval $[a, b]$ that is subdivided into n subintervals of equal length $\Delta x = (b - a)/n$ using $n + 1$ points $x_i = a + i\Delta x$:

$$x_0 = a,\ x_1 = a + \Delta x,\ \ldots,\ x_{n-1} = a + (n-1)\Delta x,\ x_n = b.$$

Then the integral $\int_a^b f(x)\,dx$ can be approximated by

$$\int_a^b f(x)\,dx \approx \sum_{i=n}^{n} f\left(\frac{x_{i-1} + x_i}{2}\right)\Delta x.$$

Numerical Approach-Midpoint Rule

- Let's divide $[a, b]$ into n segments.
- Length of one segment $h = \dfrac{(b-a)}{n}$.
- Discretized points will be given by

$$x_{(j/2)} = a + (j - 1/2)h \qquad\qquad i \leq j \leq n.$$

- Apply the mid point rule

$$\int_a^b f(x)dx \approx h.f\left(a+\frac{1}{2}h\right) + h.f\left(a+\frac{3}{2}h\right) + h.f\left(a+\frac{5}{2}h\right) + \ldots + h.f\left(a+\frac{2n-1}{2}h\right)$$
$$= h\left(.f\left(a+\frac{1}{2}h\right) + .f\left(a+\frac{3}{2}h\right) + f\left(a+\frac{5}{2}h\right) + \cdots + f\left(a+\frac{2n-1}{2}h\right)\right)$$

Mid Point code

```
function I=midpoint(func,a,b)
fa=feval(func,a);
fb=feval(func,b);
c=(a+b)/2;
fc=feval(func,c);
I=(b-a)*fc;
end
```

Command Window

```
>> func=@(x) exp(-x)+x;
>> a=1;
>> b=2;
>> midpoint(func,a,b)

ans =

  1.723130160148430
```

Composite Mid Point code

```
function I=comp_midpoint(f,a,b,n)
h=(b-a)/n;
M=0;
for i=1:n
    x=a+(i-1/2)*h;
    fx=feval(f,x);
    M=M+h*fx;
end
I=M
end
```

```
f=@(x) exp(-x)+x;
n=10;
a=1;
b=8;
comp_midpoint(f,a,b,n)
```

Trapezoid Rule

Let $f(x)$ be defined on a closed interval $[a, b]$ that is subdivided into n subintervals of equal length $\Delta x = (b - a)/n$ using $n + 1$ points $x_i = a + i\Delta x$:

$$x_0 = a,\ x_1 = a + \Delta x,\ \ldots,\ x_{n-1} = a + (n-1)\Delta x,\ x_n = b.$$

Then the integral $\int_a^b f(x)\,dx$ can be approximated by

$$\int_a^b f(x)\,dx \approx \frac{\Delta x}{2}(f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)).$$

Trapezoid Rule

```
function I=trap(f,a,b)
fa=feval(f,a);
fb=feval(f,b);
I=((b-a)/2)*(fa+fb)
end
```

```
f=@(x) exp(-x)+x;
n=10;
a=1;
b=2;
trap(f,a,b)
```

Numerical Approach-Trapezoidal Rule

- Let's divide $[a, b]$ into n segments.
- Length of one segment $h = \dfrac{(b-a)}{n}$.
- Discretized points will be given by

$$x_{(j)} = a + j * h \qquad\qquad i \leq j \leq n.$$

- Apply the Trapezoidal rule for each trapezoid:

$$\int_a^b f(x)dx \approx \frac{h}{2}\big(f(a)+f(a+h)\big)+\frac{h}{2}\big(f(a+h)+f(a+2h)\big)+\cdots+\frac{h}{2}\big(f(a+(n-1)h)+f(a+nh)\big)$$

$$=\frac{h}{2}\left(f(a)+2\sum_{i=1}^{n-1}f(a+h*i)+f(a+nh)\right)$$

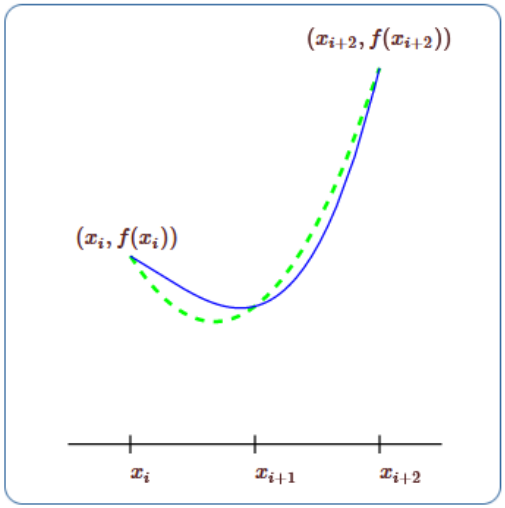
$$=\frac{h}{2}\left(f(a)+2\sum_{i=1}^{n-1}f(a+h*i)+f(b)\right)$$

Composite Trapezoid Rule

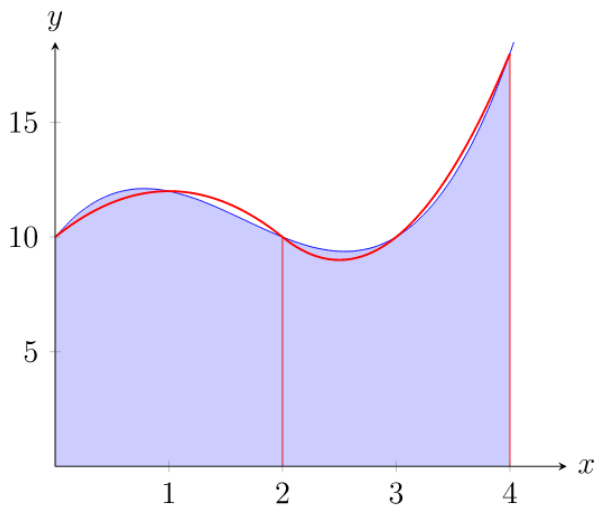
```
function I=compTrap(f,a,b,n)
h=(b-a)/n;
T=f(a);

for i=1:n-1
    x(i)=a+h*i;
    T=T+2*f(x(i));
end
I=(h/2)*(T+f(b));

end
```

```
f=@(x) exp(-x)+x;
n=10;
a=1;
b=5;
compTrap(f,a,b,n)
```

Simpson's 1/3 Rule



A parabola (dashed) approximating a curve (solid)



Approximating an area with parabolas

Simpson's 1/3 Rule

Let $f(x)$ be defined on a closed interval $[a, b]$ that is subdivided into n even subintervals of equal length $\Delta x = (b - a)/n$ using $n + 1$ points $x_i = a + i\Delta x$:

$$x_0 = a,\ x_1 = a + \Delta x,\ \ldots,\ x_{n-1} = a + (n-1)\Delta x,\ x_n = b.$$

Then the integral $\int_a^b f(x)\,dx$ can be approximated by

$$\int_a^b f(x)\,dx \approx \frac{\Delta x}{3}(f(x_0)+4f(x_1)+2f(x_2)+4f(x_3)+\cdots+4f(x_{n-1})+f(x_n)).$$

Numerical Approach-Simpson's 1/3 Rule

- Let's divide $[a, b]$ into n segments.
- Length of one segment $h = \dfrac{(b-a)}{n}$.
- Discretized points will be given by

$$x_{(j)} = a + j * h \qquad\qquad i \leq j \leq n.$$

- Apply the Simpson's 1/3 Rule for each segment of three points:

$$\int_a^b f(x)dx \approx \frac{h}{3}(f(a) + 4f(x_1) + f(x_2)) + \frac{h}{3}(f(x_2) + 4f(x_3) + f(x_4)) + \cdots + \frac{h}{2}\big(f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\big) \quad)$$

$$= \frac{h}{3}(f(a) + 4(f(x_1) + f(3) + ..) + 2(f(x_2) + f(x_4) + ..) + f(b))$$

$$= \frac{h}{3}(f(a) + \textstyle\sum_{i-odd}^{n-1} f(x_i) + \sum_{i-even}^{n-2} f(x_i) + f(b))$$

Simpson's 1/3 Rule

```
function I=simp(f,a,b)
fa=feval(f,a);
fb=feval(f,b);
c=(a+b)/2;
fc=feval(f,c);
I=((b-a)/(2*3))*(fa+4*fc+fb)

end
```

```
f=@(x) exp(-x)+x;

a=1;
b=3;
simp(f,a,b)
```

Composite Simpson's 1/3 Rule

```
function I=compsimp(f,a,b,n)
h=(b-a)/n;
S=feval(f,a);
for i=1:2:n-1
    x(i)=a+i*h;
    S=S+4*feval(f,x(i));
end
for i=2:2:n-2
    x(i)=a+i*h
    S=S+2*feval(f,x(i));
end
I=(h/3)*(S+feval(f,b))
end
```

```
f=@(x) exp(-x)+x;
n=10;
a=1;
b=5;
compsimp(f,a,b,n)
```